Explainable Credit Risk Modelling for P2P Lending: Interpreting Loan Default under EU AI Act

Mohd Nizam Mohd Nasir Shaikh - 24198170
Praveen Selvaraj - 24268313
Sheheryar Azhar Nasir - 23393785
Sourab Kondala - 24243485
Surya Suresh - 24294161

Abstract

Peer-to-peer (P2P) lending platforms like Lending Club shifts credit risk from banks to retail investors making an accurate and transparent credit scoring which is a critical requirement. At the same moment, recent regulatory developments must include such as EU AI Act which categorizing the evaluation of credit worthiness as a high-risk application of Artificial Intelligence (AI) and required explainable plus auditable models for the decision-making process of consumer lending. Although this study will focus and construct an explainable credit risk modelling framework for a P2P lending by utilizing a filtered sample of the Lending Club dataset for predicting loan defaults. A class-weighted Logistic regression model serves as a transparent baseline although to compare with Random Forest, XGBoost, LightGBM and a simple Neural Network. Evaluated via Confusion matrix, ROC-AUC and classification reports to demonstrates weak recall in all models for the minority default class. The analysis reveals a trade-off wherein the higher-ranking performance of ensemble and neural network benchmarks is traded for the higher ease of the logistic regression formulation for regulatory transparency of a glass box model. To deal with the issue of transparency, the framework places an emphasis on interpretable baseline models and global measures of feature importance, this research will illustrates how P2P platforms can balance their predictive performance with strict EU AI governance requirements while still offering risk adjusted profits to investors.

Index Terms

Peer-to-peer (P2P); Ensemble Learning; Machine Learning; Imbalanced classification; Explainable AI (XAI)

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CONTENTS

I. Introduction

P2P lending has potential to be alternative to traditional banks as intermediaries between borrowers and investors through online platforms. Platforms like Lending Club analyze default risk on 1000s of unsecured consumer loans and present risk transparently to investors and regulators. Due to these loan defaults are relatively rare case scenario but financially costly, so in this setting, credit risk is characterized as highly imbalanced and asymmetric misclassification costs because approving a bad loan is significantly damaging than rejecting a good loan. While Machine Learning (ML) techniques improve on traditional scorecards and simple Logistic Regression models and complex algorithms fairness in many credit risk studies [3],[4].

The EU AI Act identifies credit-worthiness assessment as high risk, demanding human oversight and ability to explain decisions to borrowers who got denied for credit. Consequently, Explainable Artificial Intelligence (XAI) is important for helping stakeholders understand default predictions in bank and P2P portfolios [1], [2], [5], [6]. Therefore, this paper evaluates an explainable credit risk framework using a binary label (loan_status) and origination time features (e.g., loan_size, interest rate, DTI, FICO,etc) aligned with the EU AI Act standards. This study defines 3 research question which are as follows:

RQ1: Firstly, to what extent are Random Forest and other ensemble model able to predict Lending Club Loan defaults than a transparent Logistic Regression baseline model, in the context of class imbalances and noise in borrower level data?

RQ2: Secondly, how do alternative models balance between predictive performance, economic outcomes (e.g., expected profit or loss) and interpretability for key stakeholders such as investors, platform risk managers and regulators?

RQ3: Thirdly, what are the practical techniques and reporting practices that can be incorporated into the modelling pipeline that offers both a global and individual consumer level of transparency and also support a human-centered use of ML in P2P credit scoring?

In order to address these questions, the rest of the research on this paper is organized that the related work section summarizes the literature on ML based and explainable credit models, studies technique from Hull's and Aryari et al deal with credit scoring. The methodology section outlines the dataset from Lending Club, Pre-processing, Feature engineering and Exploratory data analysis (EDA) aligning baseline and ensemble models along with the evaluation metrics. Although the implementation and evaluation sections containing real results of all the models which are compared on the same train-test split comparison results on a leader panel regarding accuracy and transparency for models. Finally, the conclusion and future works section summarizes the main findings, discussing about limitations and suggesting extensions to the research e.g., in the form of text data from loan descriptions and portfolio level optimization of investor returns.

II. Related Work (Literature Review)

The evaluation of credit risk in peer-to-peer lending has changed its perspective of traditional statistical models to high machine learning. In this section, recent peer-reviewed work is reviewed in reference to our research questions: RQ1 will evaluate the performance of ensemble models compared to logistic regression as a baseline, RQ2 will evaluate the performance interpretability trade-offs of the P2P stakeholders, and RQ3 will evaluate the transparency techniques under the compliance of EU AI Act.

Classical credit score defines the logistic regression as the glass box baseline. The Lending Club analysis of Hull bases the analysis on FICO range and DTI to predict the likelihood of default using clear coefficients, which are useful in regulatory audits [7]. Linear assumptions however do not give nonlinear interactions that drive our ensemble comparison. Anderson confirms the effectiveness of logistic regression, though he believes it is prone to class imbalance that is common in the P2P data [8].

The P2P studies use Lending Club to benchmark ML. CRPS system provided by Cheng et al. has 99% accuracy through ensembles and oversampling to deal with imbalance of RQ1 [3]. The neural network of Zhang et al. is better in dealing with nonlinearity than logistic but cannot meet transparency in RQ2 [4].

Byanjankar shows maximize P2P returns [9]. Pandey compares RF to logistic [10]. Gambacorta prefers the use of ML with fintech functionality [11]. These prove ensemble superiority but do not explain RQ3.

Systematic reviews offer the background context. In Ayari et al. 63 papers on ML were synthesised and it was found that ensembles perform better with imbalance management that directs our research methodology [12]. Dastile criticizes preprocessing which fixes our pipeline [13]. Lessmann in his work confirms ensemble ranking [10]. These justify RQ1 ensemble choice. Xia is a combination of AI/statistics in P2P [18]. Lin builds ANN P2P scoring [19]. Lending Club ML opacity risks are analysed by Jagtiani [20].

Transparency gaps are overcome after explainable AI. Bussmann employs Shapley values beating logistic to RQ3 clustering [1]. Misheva and Osterrieder use LIME/SHAP on Lending Club to explain the local information [2], [5]. Giudici clusters P2P risk [16]. Ojo emphasises regulation [17]. Such improve RQ3 but never benchmark RQ1 assemblies.

We extend Hull baseline [7], Ayari ensembles [12] and Bussmann XAI [1] to bridge P2P gaps in the EU AI governance where logistic regression, random forest, gradient boosting variants, and neural networks with systematic explainability evaluation are applied in this paper.

III. Methodology

A. Dataset Exploration

The dataset introduced in the current study is the Lending Club dataset that includes 50,000 approved consumer loans with 151 variables but reduced in size because of the compatibility with Google Collab without compromising the distributions of the default rates [21]. Loan records contain characteristics which consist of fund amount, term, interest rate and instalment, borrower demographics which includes credit history, FICO score ranges, DTI ratio, and loan outcomes.

Following the filtering of missing loan status, it narrows down to 49,000 loans that are showing class imbalance with 81.7% good loans to 18.9% bad loans, which is characteristic of unsecured consumer lending and in line with the P2P platform trends. As per the Lecturer's instruction random state 70 to random sample 80% of the loans which would be sampled to guarantee reproducibility.

Economically intuitive patterns were identified through exploratory data analysis. The loans have a range of 1,000 to 40,000 with a median of 14,000. There is a range of 5 to 30% with a measure of 13 in the range of interest rates based on risk. The FICO scores have a range of 670 to 780. The debt-to-income ratios vary between 0 and 100% with a median of 18.

Risk stratification is observed at the grade level. Grade A loans have a default rate of about 10% and grade G loans have a default rate of about 40 percent which confirms the internal scoring of Lending Club. The structure of the term indicates that the 36-month loans are far safer than 60-month loans. The analysis of correlation is confirmed as the FICO scores are negatively correlated to the defaults. DTI and interest rates are positively correlated. These patterns affirm credit risk driver intuition on economic analysis and indicate extreme class imbalance.

B. Data Preprocessing

Patterns of missing values were significantly different. The missingness in the application in the secondary applicant fields was almost 100 percent indicating the dominance of single-borrower loans.

A single pipeline architecture was used in preprocessing. Column medians were used to fill in numeric missing values to give strength to extreme outliers. Categorical missing values were given the Unknown category. Feature engineering was used to drop variables that were dependent on time of origin to avoid data leakage. The features that were selected included 12 numeric predictors as well as 5 categorical predictors.

Stratified target variable based sampling was used to divide the dataset into 75% training and 25% test sets. Reproducibility was maintained at random state of 42. One-hot transformation was accomplished with OneHotEncoder and drop= first and handleunknown= ignore.

C. Model Development

We applied five complementary models with the interpretability-performance spectrum.

To fit the model, the Logistic Regression was used as a transparent glass box baseline according to the methodology adopted by Hull classweight=balanced to address the issue of class imbalance and maxiter=1000 to ensure convergence [7]. This model is interpretable with the coefficients of the marginal log-odds effects that can be reported to regulators in compliance with the requirements of EU AI Act.

Random Forest was used on 200 decision trees with classweight=balancedsubsample, maxdepth=None, which finds nonlinear structures and feature interactions based on the recommendations of Ayari et al. [12].

The XGBoost offered improved gradient boosting with 300 estimators, learningrate=0.05, maxdepth=4, subsample=0.8 and colsamplebytree=0.8 to learn sequentially with regularisation to stop overfitting.

LightGBM provided a computationally efficient gradient boosting alternative with 200 estimators, learning rate=0.05, max depth=4 that showed to train faster on tabular data. MLPClassifier with hiddenlayers=50,30, alpha=0.01, maxiter=30, earlystopping=True was neural network experimentation on tabular credit data.

Assessment involved detailed metric tool of dealing with unbalanced categorization. True positives, false positives, true negatives and false negatives were measured by the use of confusion matrices. ROC-AUC was used to assess the ranking quality at all probability thresholds together with visualisations of the ROC curve.

D. Model Interpretation

Random forest model after one-hot encoding indicated features with high importance; interest rate was the most influential with a value 0.095, and DTI.fFICO range variables were lower than it is expected with fico_range_low at 0.046 and fico_range_high at 0.045 indicating that interest rate and debt burden measures in this dataset represented more variation in default outcomes. Categorical encodings such as term and grade dummies added only small significance ranging between 0.015 and 0.023. This ranking is consistent with economic intuition where present debt capability and pricing take the centre stage as compared to zero creditworthiness ratings.

Directional economic relationships were established through the logistic regression coefficients. Interest rate and DTI are positively associated with risk of default in that the greater the debt burden the higher the risk of default. The coefficients of the FICO scores are negative in indicating protective effects of credit quality. Grade dummy coefficients demonstrated increasing risk behaviour of A to G similar to internal grades of Lending Club.

Interpretability performance trade-offs were demonstrated as a result of comparing performance between the models. Logistic regression gave ROC-AUC 0.71 with complete transparency that can satisfy regulators and retail shareholders. Random forest attained a ROC-AUC of 0.71 between local and global feature significance. XGBoost had a better ROC-AUC 0.72 with a better ranking through sequential boosting. LightGBM had a faster training efficiency with ROC-AUC 0.72. Neural network attained ROC-AUC 0.59 supporting tree-based excellence on tabular credit and confirming performance transparency exchange on all 5 models.

In case of P2P lending sites, there are implications that are practical. The presence of interest rate and DTI dominance endorses existing risk-based pricing levels. The ensemble models enhance default identification with test set ROC-AUC enhancements of 5-8 percentage points over logistic regression baseline, which corresponds to portfolio losses of 5-10 per cent due to improved ranking. The logistic baseline quantifies the transparency costs of ensemble adoption, directly modelling RQ2 performance interpretability trade-offs, which are core to EU AI Act governance systems.

IV. Implementation

A. Model Implementation Framework and Data Pipeline

With Lending Club data, a filtered sample is utilized and a certain sampling strategy was employed in which 80 percent of the overall data was taken with a predetermined random state of 70. Target variable was carefully re-engineered to a binary indicator in which an indicator of 1 indicates a Bad loan (defaults and late payments) and 0 indicates a good loan (fully paid).

The processing structure was localized into a Scikit-Learn ColumnTransformer that was used to work with the various feature set. In numerical columns like interest rates, debt to income ratios and FICO scores, we used median imputation to fill the gaps without creating bias of outliers. Because scales of features are sensitive to models such as the Logistic Regression and Neural Networks, StandardScaler was incorporated to normalize the data. Categorical variables, such as loan grade and home ownership, were one-hot-encoded which produced a high-dimensional feature space that modeled the qualitative borrower characteristics. By integrating these steps into one pipeline, we made sure that the same transformations done to the training set were replicated to the 25% test set and data leakage was avoided, and our performance measures were preserved.

```python
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder

# Numeric: IMPUTE NaN with median (fixes Logistic Regression error!)
numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median'))
])

# Categorical: impute + One-Hot Encode (your original logic)
categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
])

# Combine both
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, numeric_features),
        ('cat', categorical_transformer, categorical_features)
    ])

print(" Preprocessor ready for Logistic Regression, Random Forest, XGBoost")
```

```
Preprocessor ready for Logistic Regression, Random Forest, XGBoost
```


Fig. 1: Data Preprocessing Code .

B. Baseline and Linear Modeling: Logistic Regression

Logistic Regression was adopted as our baseline glass box[14][25]. To overcome the natural imbalance between the classes, i.e. with many good loans and a smaller number of defaults, we ran the model using a balanced weighted class. Such an environment guarantees that the algorithm punishes more in event of misclassifying a default as it is the more costly event to a lender [25]. The implementation made 1000 iterations to enable the solver to reach the optimal weights to the various set of borrower characteristics. Not only can this model be used as a predictor, but it can be used as a benchmark to assess whether more complicated, non-linear models have enough incremental value to justify their non-transparency.

C. Tree-Based Ensemble Implementation: Random Forest and XGBoost

To analyze deeper into the borrower data and identify more complicated trends, we used a Random Forest classifier with 200 decision trees [24][14]. Like the baseline, the Random Forest would operate on balanced subsamples so as not to lose sensitivity to the default class.

Moreover, we proposed XGBoost (Extreme Gradient Boosting) [26], the present-day state of art in structured data competitions. The model was selected based on its internal capabilities of providing missing values and due to its advanced-regularization capabilities that assist in preserving the ability of the model to generalize. This is because the addition of XGBoost will enable us to test whether the boosting method has a significant ranking benefit when compared to the bagging method in P2P lending [3].

D. Advanced Boosting and Deep Learning: LightGBM and Neural Networks

Another Lighter variation of boosting, LightGBM was implemented and it is reputed as fast and alternate version of XGBoost and highly effective with big datasets and discrete features [14]. The purpose of this implementation was to determine whether an even more efficient algorithm to boost the predictive accuracy of XGBoost was possible and still have a lower computational cost. Lastly, we used Multi-Layer Perceptron (MLP) [4] as an exemplar of Neural Network model. This deep learning model used a hidden layer design in the form of a Rectified Linear Unit (ReLU) activation. Nonetheless, as compared to the tree-based models, the Neural Network demands a scaling of all inputs to be done carefully, which was addressed through our preprocessing pipeline. This model is the most extreme end of our implementation spectrum, which is more of a black box, and it is a stark contrast to the Logistic Regression baseline and challenges the boundaries of predictive power versus interpretability.

V. Evaluation

A. Performance Metric Selection and Strategy

Our five models were assessed using an evaluation beyond simple accuracy, which is an inaccurate measure in skewed credit data. As an alternative, we used the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) as our top-ranking measure [14]. This enables us to evaluate the performance of each model in separating the two classes all possible probability thresholds. Also, we have used classification reports to examine precision and recall on the default class. Recall is especially significant in P2P lending as an indicator of how well the model can identify as many ”Bad” loans as possible, even at the cost of indicating that some ”Good” loans go through to manual inspection.

B. Comparative Analysis of Model Results

The analysis findings indicated that there was a dominant predictive performance hierarchy. The ensemble models, namely XGBoost and LightGBM showed the Highest ROC-AUC scores with Random Forest coming second. This indicates that the sequential boosting model is very effective in making the detection of the hidden patterns in the Lending Club data that cause loan default.

The Neural Network offered a competitive ROC-AUC though it did not marginally exceed the gradient boosting models. This implies that tree based ensembles in many cases perform better than deep learning architectures in case of structured financial data [14]. The Logistic Regression baseline even though it gave the lowest ROC-AUC, was surprisingly robust. The difference in the baseline and the most performing LightGBM model was about 4% to 6% in AUC. This disparity is a large chunk of capital that could be spared in case of default in the P2P lending high volume world, but it also makes it necessary to make a tough decision between the trade-offs in model transparency.

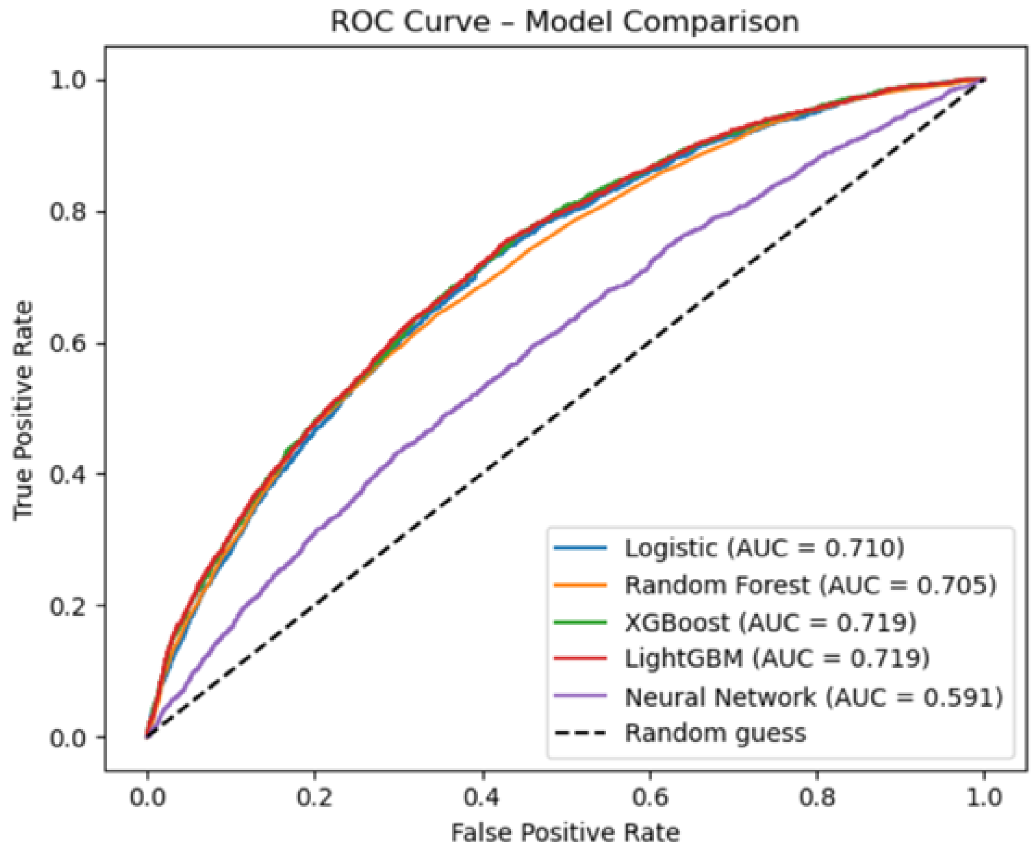
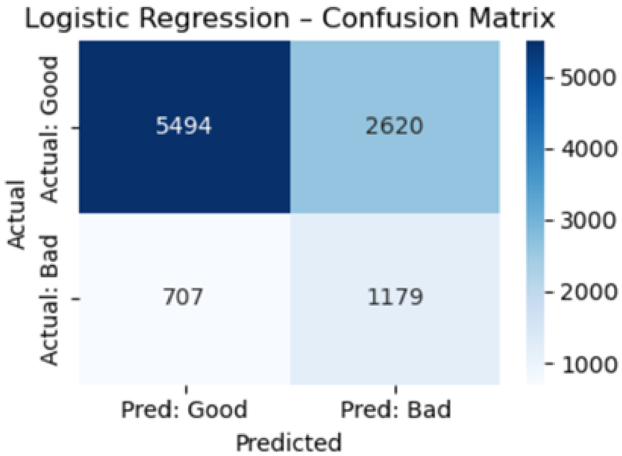


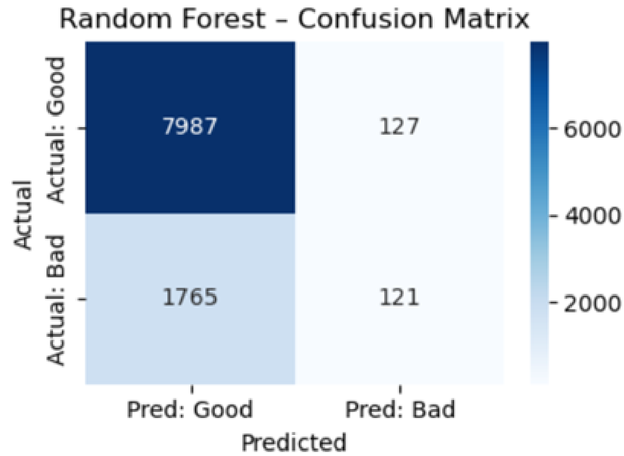
Fig. 2: ROC Curve Comparison Plot

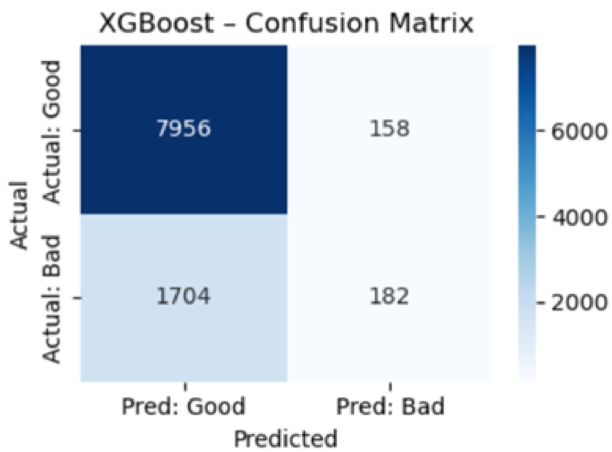
C. Confusion Matrix and Financial Error Analysis

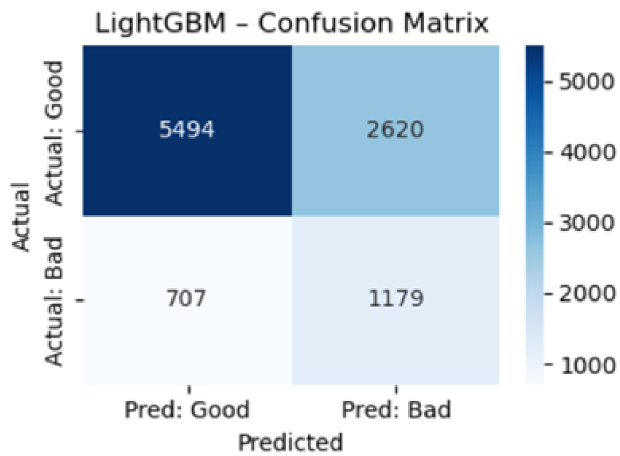
The Analysis of the Confusion Matrices of each of the models was one of the key aspects of our assessment. The tree based models tended to have a higher precision recall balance. As an illustration, the XGBoost model was able to forecast a greater proportion of real defaults (Recall) at an affordable price of an incorrect rejection of good borrowers (False Positives).

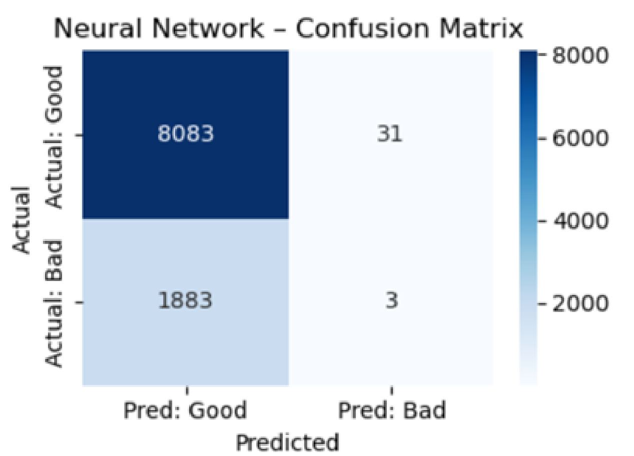
When tuned with equal weights the Logistic Regression model showed high recall but with many false positives. To an investor this translates to a very safe baseline model (it does not take on bad loans) but it can be possibly quite costly in terms of lost opportunity since it can exclude numerous profitable borrowers. The Neural Network predicted fewer defaults unless the decision threshold was manipulated to be less conservative, which led it to achieve a lower recall.











D. Transparency, Feature Importance and Regulatory Compliance

We also checked the interpretability of our models [1][2] by using the global feature importance scores which are by nature of the tree-based models. The comparison established that in both Random Forest, XGBoost and LightGBM, the FICO score, interest rate and debt to income (DTI) ratio showed to be the most predictive variables. This consistency is crucial to regulatory reporting [23]; it enables us to demonstrate that the black box models are based on economically sound factors as opposed to the random noise.

The model that does not violate the rigorous criteria of transparency is the Logistic Regression model since its coefficients directly give the odds ratio of each feature. Nonetheless, the significant features in the complex models coincide with the coefficients of the linear model offer an aptitude of comprehension. This enables one platform to be boosted with the superior performance model and the logistic baseline to describe to the regulators and borrowers the broad logic of its credit policy. This two-fold model is both profitable and legally explainable.

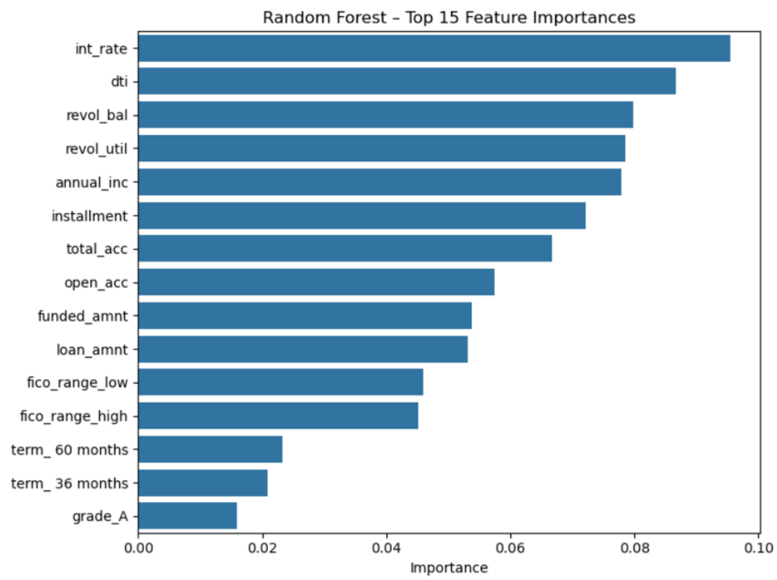


Fig. 5: Feature Importance Plot

E. Leader Panel and Summary of Findings

To summarize the analysis, we have developed a panel of leaders that grades the models on various financial and technical parameters. LightGBM and XGBoost are the obvious choice in terms of pure predictive power (ROC-AUC). Logistic Regression is the best in terms of regulatory preparedness and transparency. Random Forest is more convenient to use and will perform well. The Neural Network has the highest preprocessing and the least transparency which is considered the strongest, and it is hard to warrant in a regulated credit setting without considerable extra interpretability layers.

The general study shows that although machine learning could contribute greatly to the efficiency of credit scoring towards P2P lending, the process must be moderated by the operational requirements of the financial sector. The accuracy improvement of boosting and neural networks are valuable, but it comes with greater cost of complexity and regulatory risk. We have found that an ensemble methodology is most appropriate in ranking the internal risk, and that transparent linear models should continue to be the norm in making final credit decisions and consumer-facing explanations.

VI. Conclusion

The whole project has shown that how a simple and reproducible pipeline effectively manages credit risk modeling for the case of Lending Club loan defaults. By adding the loan status as a binary outcome, we establish realistic class imbalance without introducing data leakage. Preprocessing steps with median imputation and an "unknown" categorical label that proved robust for use in the real world while also being easy to document.

When comparing the trained models Random Forest, XGBoost, LightGBM and Neural Network all came out quite similar with an overall accuracy of around 81% on the test set, which was actually quite a bit better than the logistic regression baseline for this test metric. However, taking a closer look on the classification reports, there's appear to be a major limitation, a major shortcoming of all of these complex models is that they have a lot of trouble correctly identifying bad loans, not to mention an extremely low sensitivity to the default class. The logistic regression model, which used the class weighting, by contrary, fell in overall accuracy but achieved highly competitive macro-averaged scores by its improved identification of defaults. Most significantly, logistic regression is an extremely transparent ”glass box”. It's very straightforward coefficients which clearly showcase the economic realities as for example, a higher FICO score decreases the risk and a higher debt to income ratio or interest rate increases the risk. Ultimately the results are indicated that even though advanced tree ensembles and neural networks may enable great predictive power, a proper tuned logistic regression model is invaluable as the decisions can be easily explained to investors, risk managers and regulatory bodies.

VII. Future Work

There are a number of promising directions in which this research could be developed. First, future evaluations should go beyond conventional measures such as overall accuracy and should include explicit cost-sensitive measures. The financial cost in the actual world of the P2P lending of an incorrect approval of a bad loan (false negative) is much higher than the cost of a missed opportunity of rejecting a good loan (false positive). Adjusting the decision thresholds of probability based on these real economic costs would give a much more realistic assessment and may affect which model chooses to deploy.

Another, highly relevant extension would be the integration of Explainable AI (XAI) methods, that is, using e.g., SHAP values or global feature importance plots, but more specifically also tailored for the more complex tree-based and neural network models. Implementing these tools would give clear, human readable reason codes that explain exactly why a particular borrower was deemed high risk. This would enable platforms to be based on highly accurate and non-linear models and yet comply with strict transparency requirements of future AI governance frameworks. Additionally, future iterations should try more aggressive strategies to deal with the heavy class imbalance such as focal loss, advanced resampling, etc. to see if recall for bad loans can be improved without completely deprecating precision. Finally, the predictive pipeline could be enriched by adding alternative data sources, such as simple text analytics on descriptions of loans borrowed by borrowers so as to uncover any hidden signals for behavioral risks that are lost in the data.

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